第六章　平面电磁波

6.1 理想介质中的均匀平面波

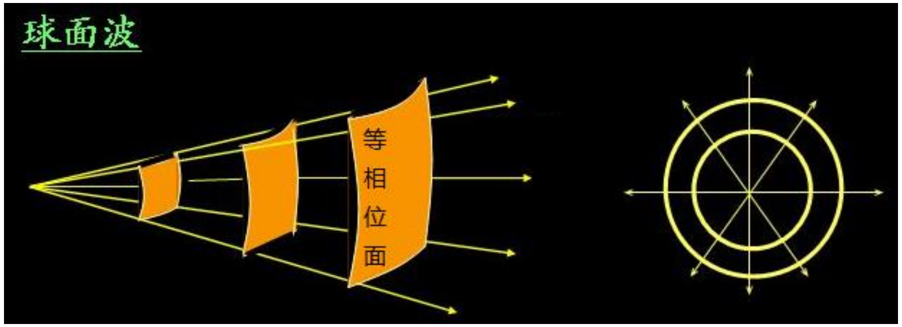
1）什么是均匀平面波?

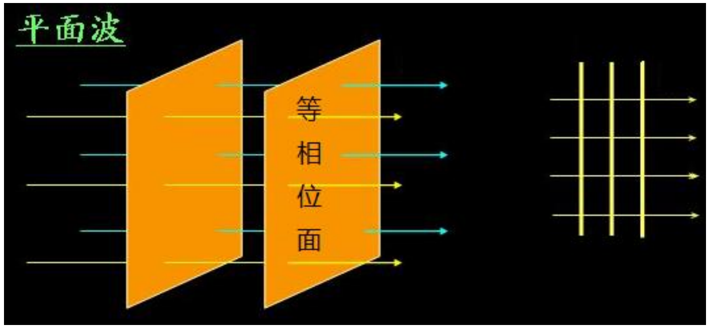
- 在同一时刻，空间相位相同的点连成的面叫等相面。

等相面为球面的波叫球面波；　　点源产生球面波；

等相面为柱面的波叫柱面波；　　无限长线源产生柱面波；

等相面为平面的波叫平面波；　　无限大面源产生平面波。





对于平面波，等相位面上各点波的振幅也相同时，是均匀平面波。否则，称为非均匀平面波。

理想介质中的波动方程：

$$\nabla^2 \overline{E}(\bar{r}) + \omega^2 \mu\varepsilon \overline{E}(\bar{r}) = \frac{1}{\varepsilon}\nabla\rho(\bar{r}) + j\omega\mu\bar{J}(\bar{r})$$

$$\nabla^2 \overline{H}(\bar{r}) + \omega^2 \mu\varepsilon \overline{H}(\bar{r}) = -\nabla\times\bar{J}(\bar{r})$$

无源区 $\rho = 0, \sigma = 0, \bar{J}(\bar{r}) = 0$

$$\nabla^2 \overline{E}(\bar{r}) + \omega^2 \mu\varepsilon \overline{E}(\bar{r}) = 0$$

$$k = \omega\sqrt{\mu\varepsilon}$$

$$\nabla^2 \overline{H}(\bar{r}) + \omega^2 \mu\varepsilon \overline{H}(\bar{r}) = 0$$

$$\nabla^2 \overline{E}(\bar{r}) + k^2 \overline{E}(\bar{r}) = 0$$

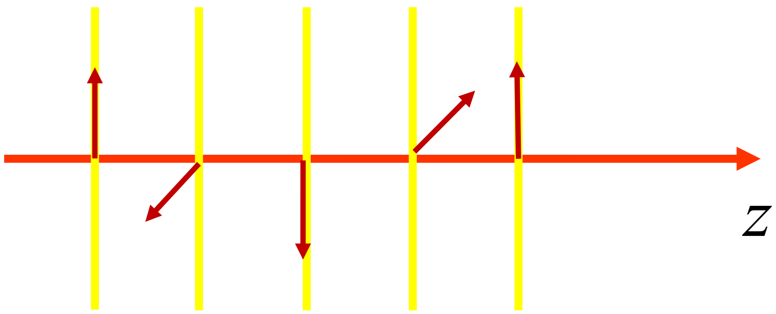
齐次矢量亥姆霍兹方程

$$\nabla^2 \overline{H}(\bar{r}) + k^2 \overline{H}(\bar{r}) = 0$$

2） 向z方向传播的均匀平面电磁波的电场

如果均匀平面电磁波向z方向传播

$$\vec{E}(\vec{r}) \longrightarrow \vec{E}(z)$$



$$\nabla^2 \vec{E}(z) + k^2 \vec{E}(z) = 0$$

电场不随x、y坐标变化，只与z坐标有关

在理想介质中的无源区：

$$\sigma = 0 \qquad \rho = 0$$

$$\nabla \cdot \vec{D} = 0$$

$$\nabla \cdot \vec{E} = 0 \qquad \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0 \qquad \frac{\partial E_z}{\partial z} = 0 \qquad E_z(z) = C \qquad E_z = 0$$

$$\vec{E}(z) \perp \hat{z} \qquad \vec{E}(z) = E_x(z)\hat{x} + E_y(z)\hat{y}$$

3） 齐次波动方程的解

在无源的理想介质中 $\nabla^2\vec{E}+k^2\vec{E}=0$ $k=\omega\sqrt{\mu\varepsilon}$

对于向z方向传播均匀平面电磁波 $\vec{E}(z)=E_x\hat{x}+E_y\hat{y}$

$$\frac{d^2E_x}{dz^2}+k^2E_x=0$$

$$\nabla^2E_x+k^2E_x=0$$

$$\nabla^2E_y+k^2E_y=0$$

$$E_x=\underline{E_{x0}^{+}e^{-jkz}}+E_{x0}^{-}e^{jkz}$$

$$E_y=E_{y0}^{+}e^{-jkz}+E_{y0}^{-}e^{jkz}$$

$$E_x^{+}=\underline{E_{x0}^{+}}e^{-jkz}$$

$$E_{x0}^{+}=\left|E_{x0}^{+}\right|e^{j\varphi_0}$$ 其复数值与波源有关

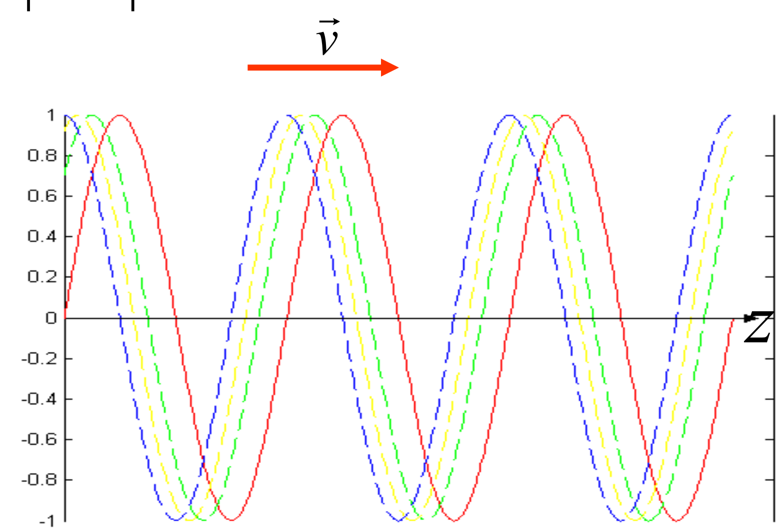
$$E_x^{+}(z,t)=\sqrt{2}\left|E_{x0}^{+}\right|\cos(\omega t-kz+\varphi_0)$$

$t=0$ 蓝

$t=T/8$ 黄

$t=3T/16$ 绿

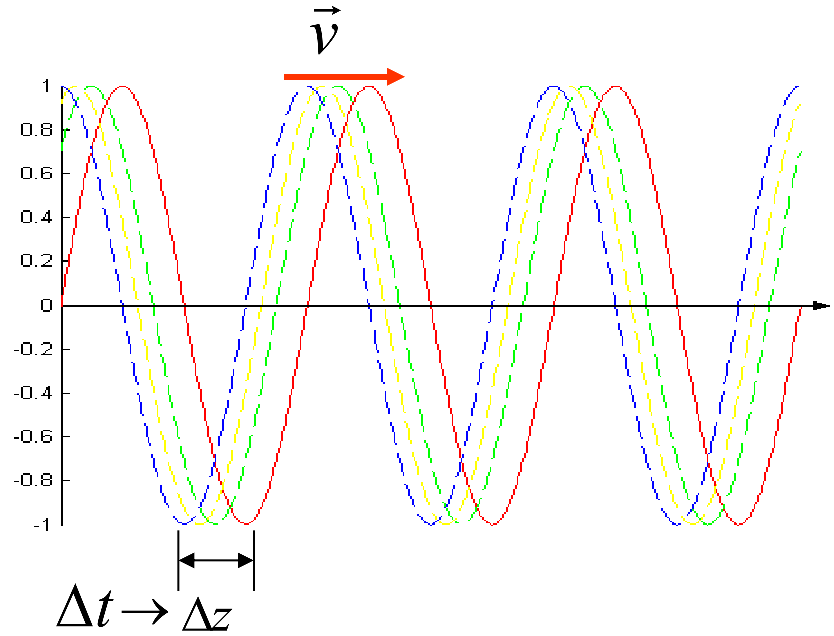
$t=T/4$ 红



4） 向z方向传播的均匀平面波

$$E_x^+ = E_{x0}^+ e^{-jkz}$$

$$E_x^+(z,t) = \sqrt{2}\left|E_{x0}^+\right|\cos(\omega t - kz + \varphi_0)$$



时间相位：ωt；

ωt变化 2π 的时间为周期 T:

$\omega T = 2\pi$ $\quad T = 2\pi / \omega$

ω表示单位时间的相位变化。单位时间内的周期数为频率：$f = \omega / 2\pi$

空间相位：kz $\quad kz$变化 2π的距离为 波长： $k\lambda = 2\pi$ $\quad \lambda = 2\pi / k$

k表示单位长度的相位变化，又叫 波数：$k = 2\pi / \lambda$ $\quad k = \omega\sqrt{\mu\varepsilon}$

相速：相位不变点移动的速度。

$$\Delta\varphi = \omega\Delta t - k\Delta z = 0$$

$$v_p = \frac{\Delta z}{\Delta t} = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}} \qquad c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \approx 3\times10^8 m/s \qquad v_p = \frac{1}{\sqrt{\mu_r\varepsilon_r\mu_0\varepsilon_0}} = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$$

$$\lambda = 2\pi / k$$

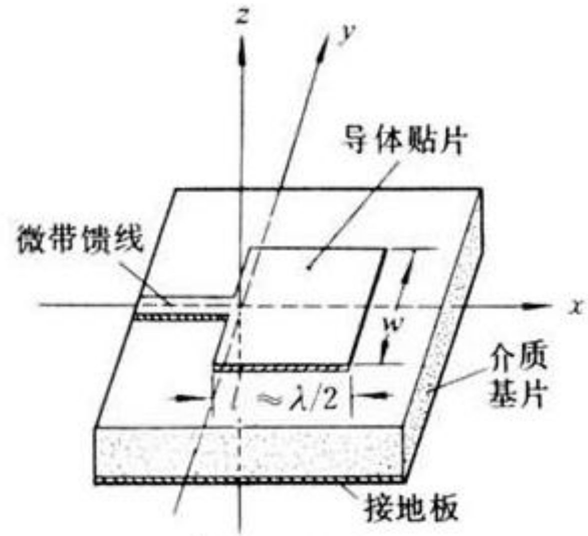
$$= \frac{2\pi}{\omega\sqrt{\varepsilon\mu}}$$

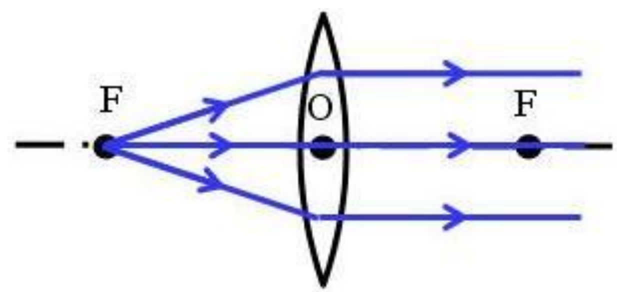
$$= \frac{2\pi}{2\pi f\sqrt{\varepsilon\mu}} = \frac{1}{f\sqrt{\varepsilon_r\mu_r}\sqrt{\varepsilon_0\mu_0}}$$

$$= \frac{c}{f\sqrt{\varepsilon_r\mu_r}}$$

$$= \frac{\lambda_0}{\sqrt{\varepsilon_r\mu_r}}$$

$$\leq \lambda_0$$





$$E_x = E_{x0}^{+} e^{-jkz} + \underline{E_{x0}^{-} e^{jkz}}$$

$$E_x^{-} = E_{x0}^{-} e^{jkz}$$ 沿负z方向传播的波

$$E_x^{-}(z,t) = \sqrt{2}\left|E_{x0}^{-}\right| \cos(\omega t + kz + \varphi'_0)$$

5) 磁场

$$\nabla \times \vec{E} = -j\omega\mu\vec{H} \qquad \vec{E} = \hat{x} E_{x0}^{+} e^{-jkz}$$

$$\vec{H} = \frac{\nabla \times \vec{E}}{-j\omega\mu}$$

$$= \frac{1}{-j\omega\mu} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ E_{x0}^{+} e^{-jkz} & 0 & 0 \end{vmatrix}$$

$$= \hat{y} \frac{k}{\omega\mu} E_{x0}^{+} e^{-jkz} \quad = \hat{y} \frac{k}{\omega\mu} E$$

波阻抗

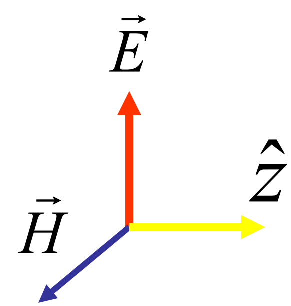
$$Z = \frac{E}{H} = \frac{\omega\mu}{k} = \sqrt{\frac{\mu}{\varepsilon}}$$

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi \approx 377\Omega$$

$$\overline{H} = \hat{y} \frac{1}{Z} E_{x0}^{+} e^{-jkz}$$

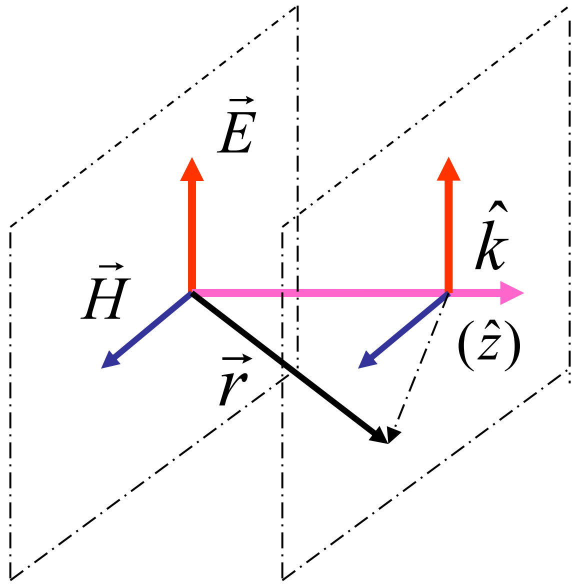
$$\vec{H} = \hat{z} \times \hat{x} E_{x0} e^{-jkz} / Z = \hat{z} \times \vec{E} / Z$$

理想介质中，均匀平面波



横电磁波 *TEM*

6）均匀平面电磁波的特性



(1) TEM 波

传播方向 $\hat{z} \rightarrow \hat{k}$

$$\vec{E} \perp \hat{k} \qquad \vec{H} \perp \hat{k} \qquad \vec{E} \perp \vec{H}$$

(2) 等相面是平面

$$e^{-jkz} \rightarrow e^{-jk\vec{r}\cdot\hat{k}}$$

$$\vec{k} = k\hat{k}$$

向k方向传播的波的空间相位因子 $e^{-j\vec{k}\cdot\vec{r}}$

$$\vec{E} = \vec{E}_0 e^{-j\vec{k}\cdot\vec{r}} = \vec{E}_0 e^{-j(k_x x + k_y y + k_z z)}$$

$$\vec{H} = \vec{H}_0 e^{-j\vec{k}\cdot\vec{r}} \qquad \vec{H}_0 = \hat{k} \times \vec{E}_0 / Z$$

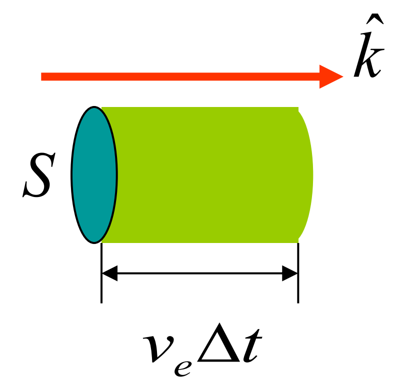
(4)　能量与功率流

$$\overline{w}_e = \frac{1}{2}\varepsilon\vec{E}\cdot\vec{E}^* = \frac{1}{2}\varepsilon|E_0|^2$$

$$\overline{w}_m = \frac{1}{2}\mu\vec{H}\cdot\vec{H}^* = \frac{1}{2}\mu|H_0|^2 = \frac{1}{2}\mu\left|\frac{E_0}{Z}\right|^2 = \overline{w}_e$$

$$\vec{S}_c = \vec{E}\times\vec{H}^* = \hat{k}|E_0|^2/Z$$

(5) 能量流动速度——能速　v_e



$$P = \iint_S \vec{S}_c\cdot d\vec{S} = S|E_0|^2/Z$$

Δt　$$\Delta W = P\Delta t = S|E_0|^2\Delta t/Z$$

$$\Delta W = (\overline{w}_e + \overline{w}_m)Sv_e\Delta t = \mu\left|\frac{E_0}{Z}\right|^2 Sv_e\Delta t$$

$$v_e = \left(\frac{S|E_0|^2}{Z}\Delta t\right)\Big/\left(\mu\left|\frac{E_0}{Z}\right|^2 S\Delta t\right) = \frac{1}{\sqrt{\mu\varepsilon}} = v_p$$

例　均匀平面电磁波，f=300MHz，在真空中沿x方向传播，电场强度方向为z向，振幅有效值为10V/m.

求:电磁波波长;写出电磁场的复数和时间(域)表达式.

解:　$f=300MHz$　　$\lambda=\frac{v_p}{f}=\frac{c}{f}=\frac{3\times10^8}{300\times10^6}=1m$　　$k=\frac{2\pi}{\lambda}=2\pi$

$$\hat{k}=\hat{x}\qquad \vec{E}_0=10\hat{z}$$

$$\vec{E}=\vec{E}_0e^{-j\vec{k}\cdot\vec{r}}=\hat{z}10e^{-j2\pi x}$$

$$\vec{E}(t)=\hat{z}\sqrt{2}\times10\cos(2\pi\times300\times10^6t-2\pi x)\quad V/m$$

$$Z=\sqrt{\frac{\mu_0}{\varepsilon_0}}=120\pi$$

$$\vec{H}=\hat{k}\times\vec{E}/Z=-\hat{y}\frac{10}{120\pi}e^{-j2\pi x}$$

$$\vec{H}(t)=-\hat{y}\frac{\sqrt{2}}{12\pi}\cos(2\pi\times300\times10^6t-2\pi x)\quad A/m$$

例：已知均匀平面波在真空中向正z方向传播，其电场强度的瞬时值为：

$$\overline{E}(z,t)=\hat{x}20\sqrt{2}\cos(6\pi\times10^{8}t-2\pi z)$$

求（1）频率及波长；（2）电场强度和磁场强度的复矢量表示式；（3）复能流密度矢量；（4）相速及能速。

解：

（1）频率： $f=\dfrac{\omega}{2\pi}=\dfrac{6\pi\times10^{8}}{2\pi}=3\times10^{8}\,Hz$

波长： $\lambda=\dfrac{2\pi}{k}=\dfrac{2\pi}{2\pi}=1m$

（2）电场强度： $\overline{E}(z)=\hat{x}20e^{-j2\pi z}\quad V/m$

磁场强度： $\overline{H}(z)=\dfrac{1}{Z_0}\hat{z}\times\overline{E}(z)=\hat{y}\dfrac{1}{6\pi}e^{-j2\pi z}\quad A/m$

（3）复能流密度： $\overline{S}_c=\overline{E}(z)\times\overline{H}^{*}(z)=\hat{z}\dfrac{10}{3\pi}\;W/m^{2}$

（4）相速及能速： $v_p=v_e=\dfrac{\omega}{k}=3\times10^{8}m/s$

6.2 导电媒质中的均匀平面电磁波 $\sigma \neq 0$

1) 导电媒质中的场方程和波动方程

$$\nabla \times \vec{H} = \vec{J} + j\omega\varepsilon\vec{E}$$

$$\nabla \times \vec{H} = \sigma\vec{E} + j\omega\varepsilon\vec{E}$$

$$\nabla \times \vec{H} = j\omega(\varepsilon + \frac{\sigma}{j\omega})\vec{E}$$

$$\varepsilon_c = \varepsilon + \frac{\sigma}{j\omega} = \varepsilon(1 - j\frac{\sigma}{\omega\varepsilon})$$

$$\nabla \times \vec{H} = j\omega\varepsilon_c\vec{E}$$

$$\nabla \times \vec{H} = j\omega\varepsilon\vec{E}$$

$$\nabla^2\vec{E} + k_c{}^2\vec{E} = 0$$

$$\nabla^2\vec{E} + k^2\vec{E} = 0$$

$$k_c{}^2 = \omega^2\mu\varepsilon_c$$

$$k^2 = \omega^2\mu\varepsilon$$

$$k \to k_c$$

$$k' = \omega\sqrt{\frac{\mu\varepsilon}{2}(\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2}+1)}$$

$$k_c = k' - jk''$$

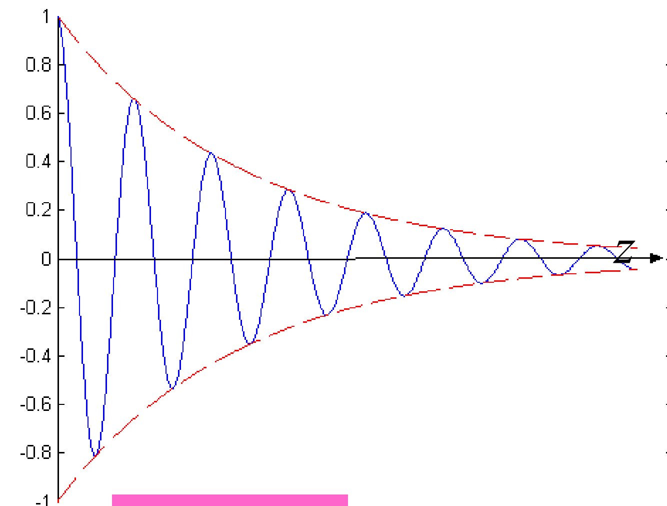
$$k'' = \omega\sqrt{\frac{\mu\varepsilon}{2}(\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2}-1)}$$

2) 导电媒质中的均匀平面电磁波

$$E_x^+ = E_{x0}^+ e^{-jk_c z} \qquad k_c = k' - jk'' \qquad -jk_c = -j(k' - jk'') = -k'' - jk'$$

$$E_x^+ = E_{x0}^+ e^{-k''z} e^{-jk'z}$$

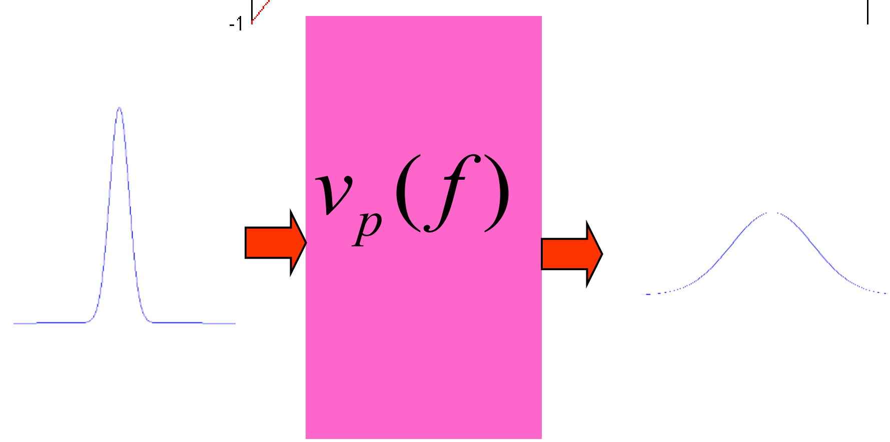
$$E_x^+(t) = \sqrt{2}\left|E_{x0}^+\right| e^{-k''z} \cos(\omega t - k'z + \varphi_0)$$



$$v'_p = \frac{\omega}{k'} = \sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right)}$$

$$\lambda' = \frac{2\pi}{k'}$$

$$Z_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon\left(1 - j\dfrac{\sigma}{\omega\varepsilon}\right)}} = \left|Z_c\right| e^{j\alpha}$$



$$\vec{H} = \hat{k} \times \vec{E} / Z_c = \hat{k} \times \vec{E} e^{-j\alpha} / \left|Z_c\right|$$

$$\vec{S}_c = \vec{E} \times \vec{H}^* = \hat{k} \frac{\left|E_{x0}^+\right|^2}{\left|Z_c\right|} e^{-2k''z} e^{j\alpha}$$

损耗角正切:

$$\tan\delta_c = \frac{\sigma}{\omega\varepsilon}$$ 传导电流密度/位移电流密度 $\frac{\sigma}{\omega\varepsilon} << 1$ $\frac{\sigma}{\omega\varepsilon} >> 1$

3) $\frac{\sigma}{\omega\varepsilon} << 1$ 低损耗的媒质 $\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2} \approx 1+\frac{1}{2}(\frac{\sigma}{\omega\varepsilon})^2$

$$k' = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2}+1\right)} \approx \omega\sqrt{\mu\varepsilon} = k$$

$$k'' = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2}-1\right)} \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$$

$$Z_c = \sqrt{\frac{\mu}{\varepsilon(1-j\frac{\sigma}{\omega\varepsilon})}} \approx \sqrt{\frac{\mu}{\varepsilon}} = Z$$

$$E_x^+ = E_{x0}^+ e^{-k''z} e^{-jkz}$$

$$\vec{H} = \hat{k}\times\vec{E}/Z \qquad v_p' = \frac{\omega}{k} \qquad \lambda' = \frac{2\pi}{k}$$

4) $\frac{\sigma}{\omega\varepsilon} >> 1$ $\quad \sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2} \approx \frac{\sigma}{\omega\varepsilon}$

$$k' = \omega\sqrt{\frac{\mu\varepsilon}{2}(\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2}+1)} \approx \sqrt{\frac{\omega\mu\sigma}{2}} \qquad k'' = \omega\sqrt{\frac{\mu\varepsilon}{2}(\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2}-1)} \approx \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$k' \approx k'' \approx \sqrt{\pi f\mu\sigma} \qquad Z_c = \sqrt{\frac{\mu}{\varepsilon(1-j\frac{\sigma}{\omega\varepsilon})}} \approx \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}e^{j\pi/2}} = \sqrt{\frac{\omega\mu}{\sigma}}e^{j\pi/4}$$

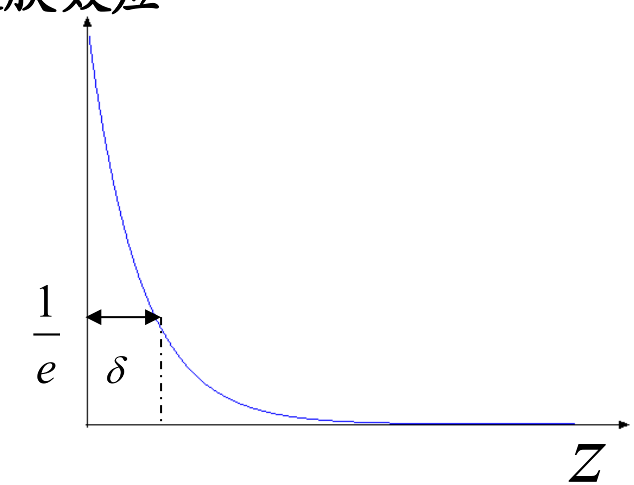
$$Z_c \approx \sqrt{\frac{\pi f\mu}{\sigma}}(1+j)$$

$$E \to e^{-k''z} \qquad f,\sigma\uparrow \to k''\uparrow \to e^{-k''z}\downarrow$$

趋肤效应

趋肤厚度 δ $\quad e^{-k''\delta} = e^{-1}$

$$\delta = \frac{1}{k''} = \frac{1}{\sqrt{\pi f\mu\sigma}}$$



f	$10kHz$	$1MHz$	$10GHz$
铜 δ	$0.6mm$	$0.06mm$	$0.6\mu m$

趋肤效应使电场、磁场仅存在于导体表面，
从而使电流主要集中在良导体表面薄层中

$$\vec{J}_s = \hat{n}\times\vec{H}\Big|_S$$

例：海水的电参数为 $\sigma = 4S/m$ $\varepsilon_r = 81$ $\mu_r = 1$

计算当频率分别为$f = 300Hz, 300kHz, 300MHz$

对应的波长和衰减4.34dB(*1/e*)对应的深度。

解：

$$\frac{\sigma}{\omega\varepsilon} = \frac{4}{2\pi f \times 81 \times 8.854 \times 10^{-12}} = 8.88 \times 10^8 / f = \begin{cases} 2.96 \times 10^6, f = 300Hz \\ 2.96 \times 10^3, f = 300kHz \\ 2.96, f = 300MHz \end{cases}$$

$$\frac{\sigma}{\omega\varepsilon} >> 1$$

$$k' \approx k'' \approx \sqrt{\pi f \mu \sigma} = \sqrt{f}\sqrt{\pi\mu\sigma} \approx 4 \times 10^{-3}\sqrt{f} = \begin{cases} 0.0693, f = 300Hz \\ 2.18, f = 300kHz \\ 69.3, f = 300MHz \end{cases}$$

$$\lambda' = \frac{2\pi}{k'} = \begin{cases} 90.6m, f = 300Hz \\ 2.88, f = 300kHz \\ 0.091, f = 300MHz \end{cases}$$

$$d = \frac{1}{k''} = \begin{cases} 14.43m, f = 300Hz \\ 0.459m, f = 300kHz \\ 0.014m, f = 300MHz \end{cases}$$

6.3 频带传输中的群速

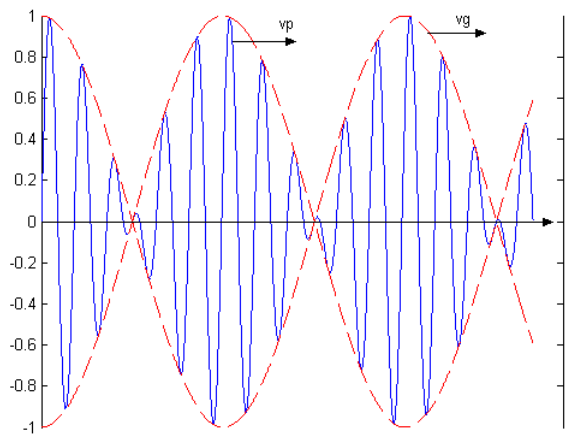
- 设两个频率相近沿z轴传播的电磁波信号的表达式分别为:

$$A_1(z,t) = A_0\cos(\omega_1 t - k_1 z) \qquad A_2(z,t) = A_0\cos(\omega_2 t - k_2 z)$$

合成波为: $A(z,t) = A_1 + A_2$

$$= 2A_0\cos\left(\frac{\omega_2-\omega_1}{2}t - \frac{k_2-k_1}{2}z\right)\cos\left(\frac{\omega_2+\omega_1}{2}t - \frac{k_2+k_1}{2}z\right)$$

$$A(z,t) = 2A_0\cos(\Delta\omega t - \Delta k z)\cos(\omega_0 t - k_0 z)$$



调幅的正弦波，载波的相速度为:

$$\upsilon_p = \frac{\omega_0}{k_0}$$

群速:调幅包络的相速度

$$\upsilon_g = \frac{\Delta\omega}{\Delta k}$$

二、色散媒质中的群速

对窄带信号，k可在载频附近泰勒展开：

$$k(\omega)=k(\omega_0)+\left(\frac{dk}{d\omega}\right)_{\omega=\omega_0}(\omega-\omega_0)+\cdots$$ 忽略无穷小量

$$\upsilon_g=\frac{\Delta\omega}{\Delta k}\approx\left(\frac{dk}{d\omega}\right)_{\omega_0}^{-1}$$ $$k=\frac{\omega}{\upsilon_p}$$

$$\frac{1}{\upsilon_g}=\left(\frac{dk}{d\omega}\right)_{\omega_0}=\left[\frac{d}{d\omega}\left(\frac{\omega}{\upsilon_p}\right)\right]_{\omega_0}=\frac{1}{\upsilon_p}\left[1-\frac{\omega}{\upsilon_p}\left(\frac{d\upsilon_p}{d\omega}\right)_{\omega_0}\right]$$

$$\upsilon_g=\frac{\upsilon_p}{1-\frac{\omega}{\upsilon_p}\left(\frac{d\upsilon_p}{d\omega}\right)_{\omega_0}}$$

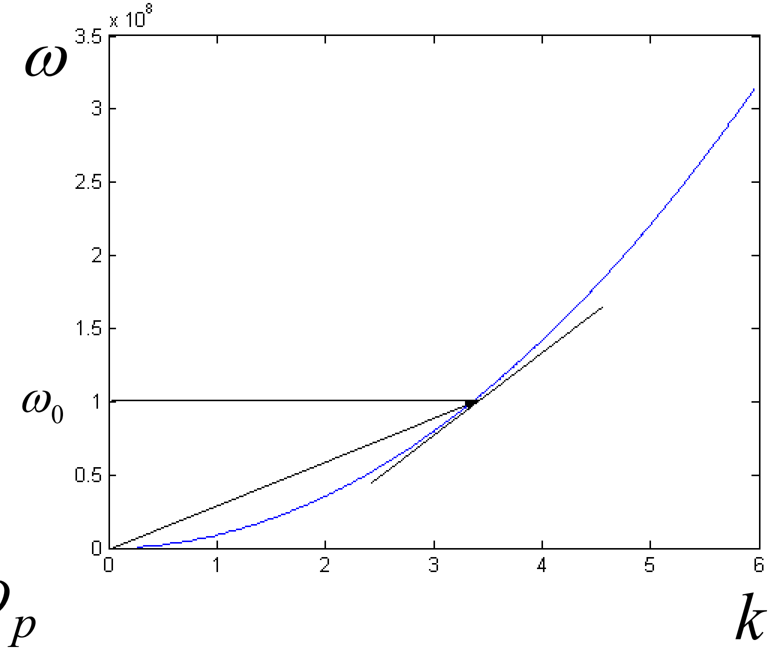
$\frac{d\upsilon_p}{d\omega}\langle 0$ $\upsilon_g\langle\upsilon_p$ 正常色散

$\frac{d\upsilon_p}{d\omega}\rangle 0$ $\upsilon_g\rangle\upsilon_p$ 非正常色散

$\frac{d\upsilon_p}{d\omega}=0$ $\upsilon_g=\upsilon_p$ 无色散

3) 相速与群速的关系

$$v_p = \frac{\omega_0}{k_0} = \frac{\omega}{k}\bigg|_{\omega_0} \qquad v_g = \frac{d\omega}{dk}\bigg|_{\omega_0}$$



对理想介质：

$$k = \omega\sqrt{\mu\varepsilon} \qquad \upsilon_g = 1/\sqrt{\mu\varepsilon} = \upsilon_p$$

对导电媒质：

$$k' = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right)} \qquad \upsilon_g = \frac{\upsilon_p}{1-\dfrac{\omega}{\upsilon_p}\left(\dfrac{d\upsilon_p}{d\omega}\right)_{\omega_0}}$$